

Bladeless Wind Turbine Numerical Analysis

7ENT2059-0901-2025 – CFD and Applications

Group 10

Name: Rupeshwaran Paranthaman
Student ID: 23069976

Name: Mohd Maaz Khan
Student Id: 24156323

Name: Abdhul Rouf Chirackal Amanulla
Student ID: 23098816

Name: Muhammad Ammar Ahmed
Student ID: 24144928

Name: Rehan Rahmat Jamal Siddiqui
Student ID: 24118823

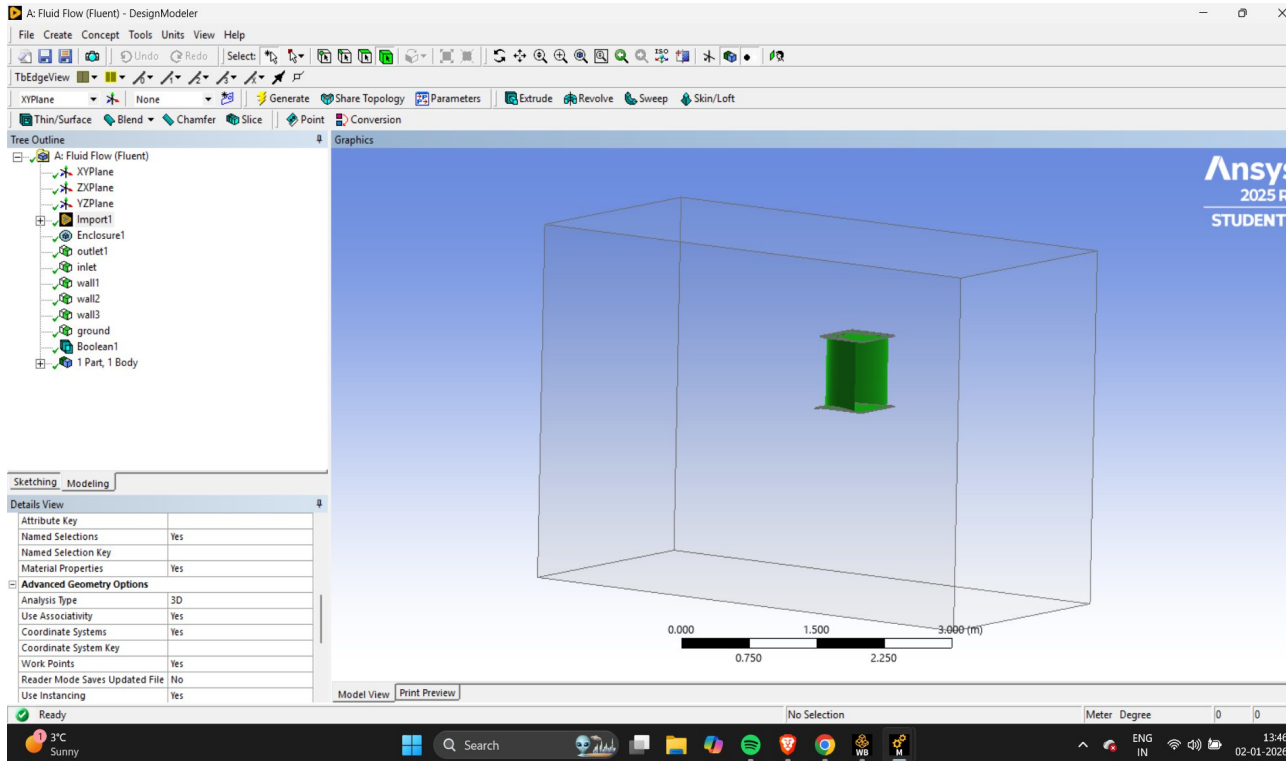


INTRODUCTION

- To effectively simulate the external airflow around the bladeless wind turbine and reduce boundary-layer interference, we created an enclosure, which serves as our computational domain.
- We clearly defined the boundaries of this enclosure such as the velocity inlet, pressure outlet, and walls to ensure that we applied the right boundary conditions in ANSYS Fluent.
- A surface mesh was generated on both the turbine and the enclosure. This mesh helps us accurately capture the flow behavior and how it interacts with surfaces.
- We paid close attention to mesh quality parameters to ensure that our simulations would converge smoothly and yield reliable numerical results.
- Following the professor's instructions for Group 10, we set a uniform inlet velocity of 44 m/s at the velocity inlet to maintain consistency in our analysis.
- Our analysis focused on examining the velocity contours, pressure distribution, and flow patterns around the bladeless wind turbine, providing us with valuable insights into its performance.



ENCLOSURE

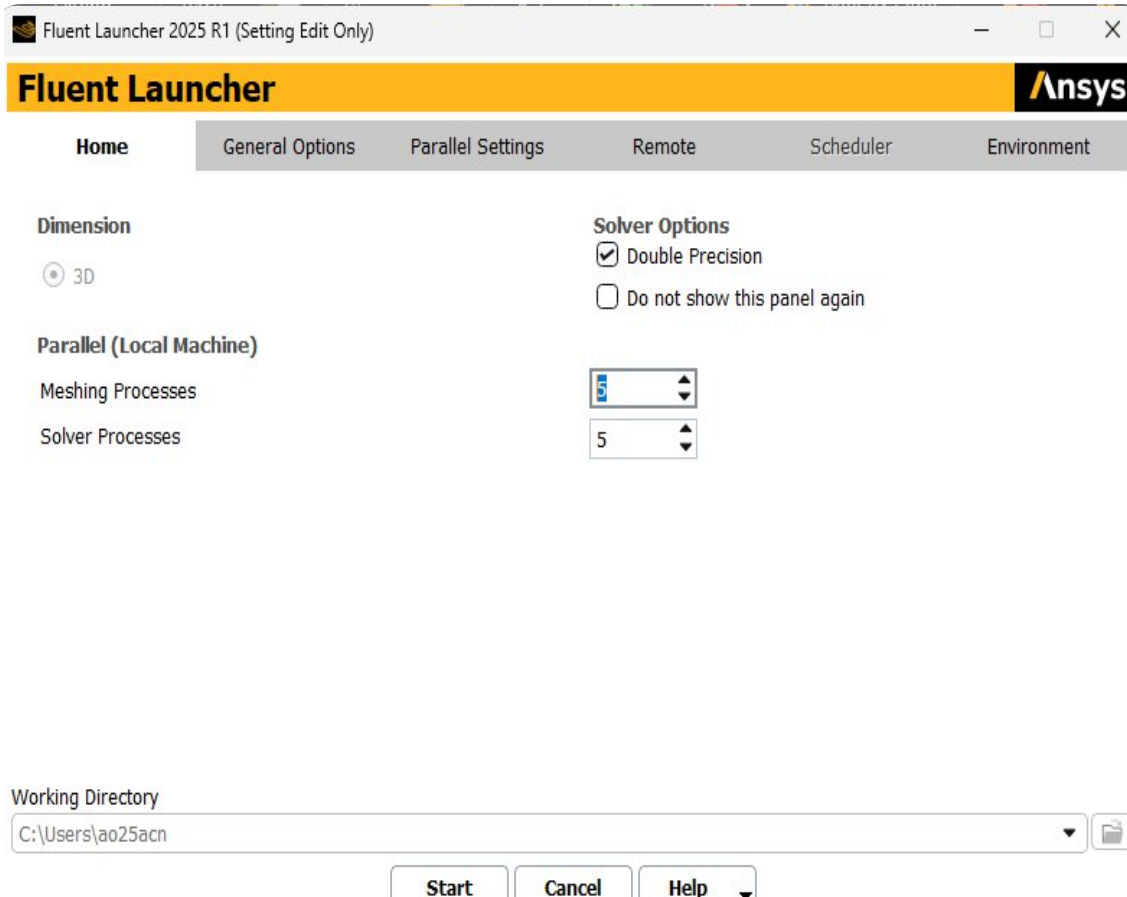


- Upstream (1500 mm): This is to make sure that the inlet flow will be 7 m/s and 15 m/s it will be completely developed before striking the turbine engine.
- Downstream (5000mm): It allows sufficient room to take into account the entire length of the low-pressure suction wake.
- Clearance (1000 mm): Dimensions are to be designed to eliminate the blockage effects due to the artificial acceleration of the air by the walls.
- Validation: The high volume of the test system will precludes interference of the local pressure gradients with the boundary conditions.



Solver Settings

- We used the Double Precision 3D solver to remove round-off errors and obtained high-fidelity pressure output results.
- We meshed and solved 5 parallel processes to ensure that the computational time is greatly reduced.
- This engineering arrangement is necessary to allow the steady-state force plateaus to be accurate in engineering analysis design.





Mesh Refinement and Local Sizing

Workflow

✓ Import Geometry

⚠ Add Local Sizing

Would you like to add local sizing?

Add Local Sizing

Name

Size Control Type

Target Mesh Size [m]

Growth Rate

Select By

Labels [1/6]

outlet

turbine

walls

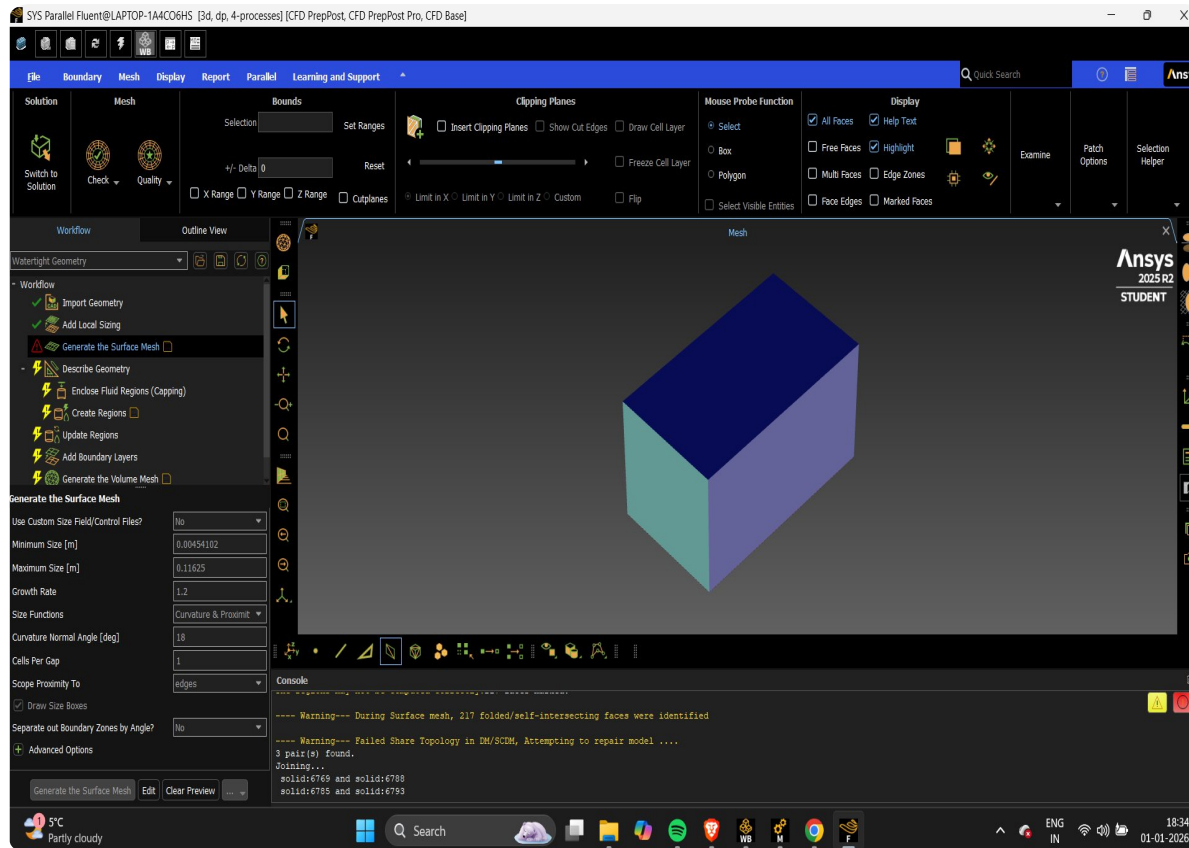
☒ Draw Size Boxes

- **3mm Surface Mesh:** This fine resolution ensures that pressure gradients and areas of flow separation are accurately represented without any numerical errors.
- **1.2 Growth Rate:** This option is selected to ensure mathematical stability during the transition to the far-field, maximizing solution time while maintaining accuracy.



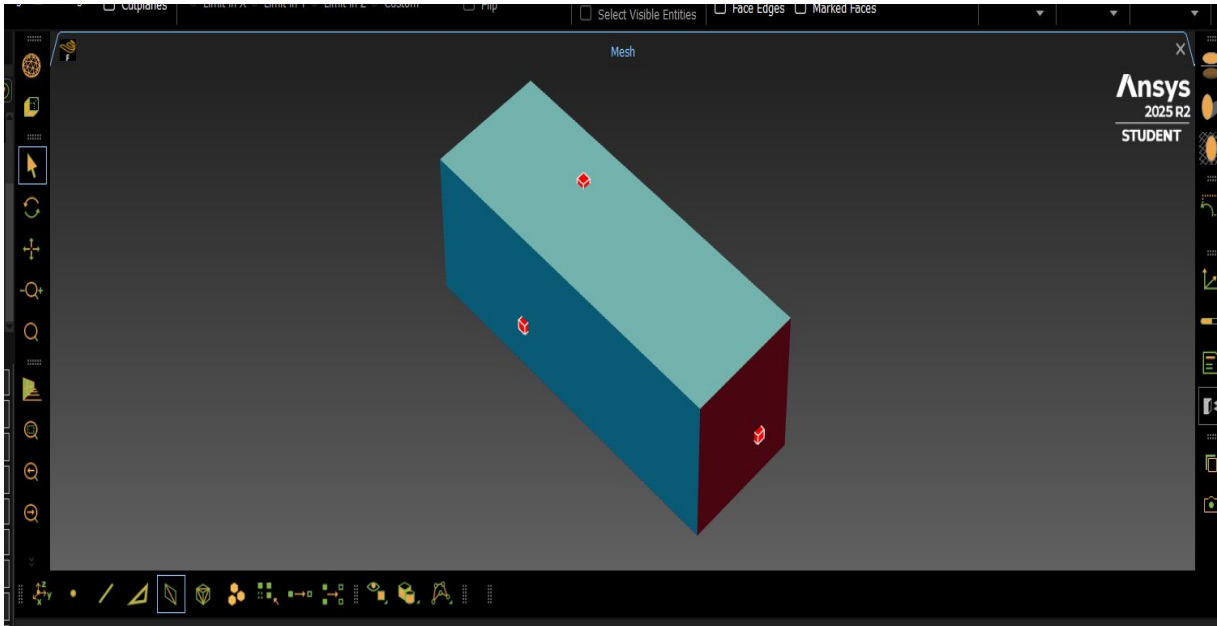
Computational Domain and Mesh Setup

- The computational domain we're working with is a rectangular enclosure, created using the Watertight Geometry workflow in ANSYS Fluent. This setup represents the fluid area around the turbine and ensures that there's enough length both upstream and downstream of it. The upstream section helps to fully develop the flow at the inlet, while the downstream part captures the wake effects that occur behind the turbine. We've also made sure to keep enough space between the model and the walls to reduce any blockage effects. For the mesh, we have created a surface mesh using sizing functions based on curvature and proximity. This method helps us maintain accuracy in the critical areas while also being efficient with our computational resources. Before we move on to volume meshing, we conduct checks and automatic repairs to ensure that the mesh quality is up to par.





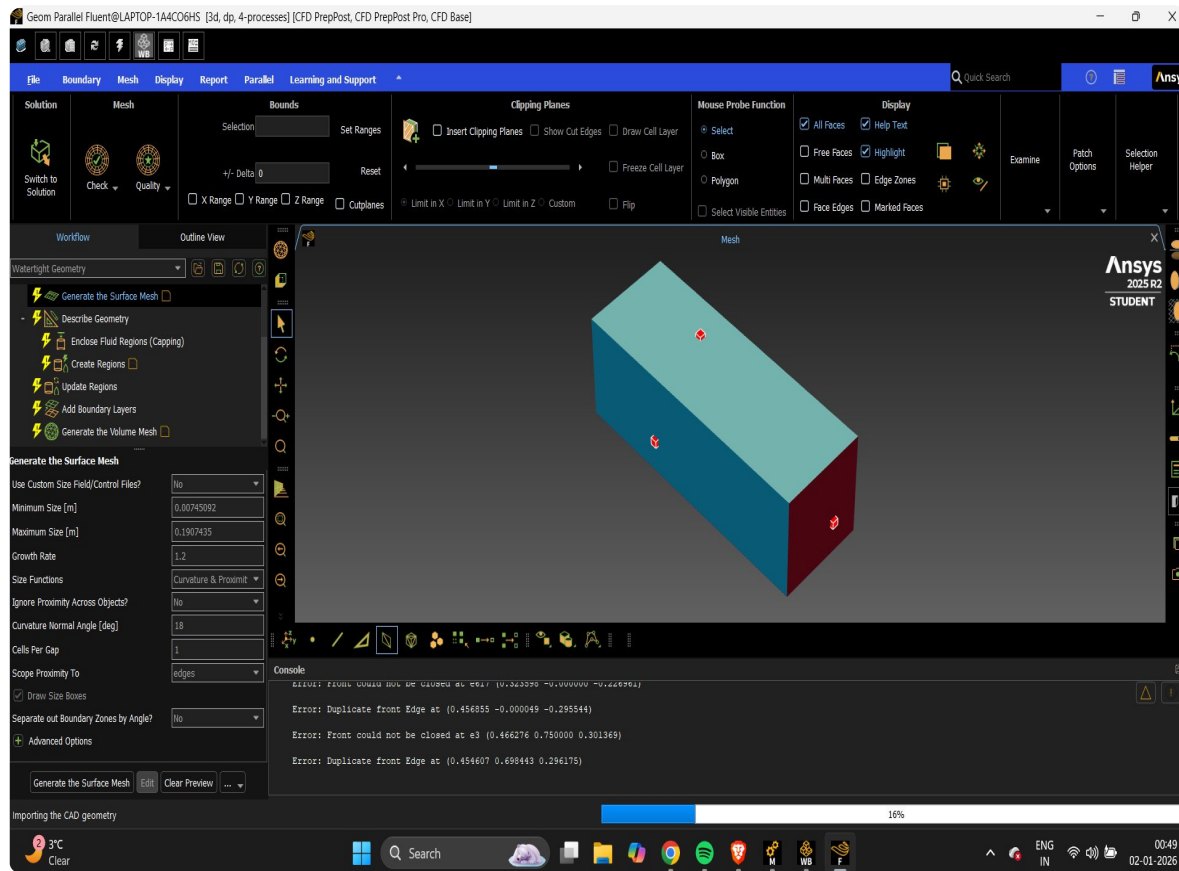
Surface Mesh Generation of the Computational Domain



- The figure displays the surface mesh created for the computational domain using ANSYS Meshing. We had applied an unstructured triangular mesh to break down the domain surfaces. To get the best element distribution and quality, we have used adaptive sizing. This method helps us achieve good resolution without driving up computational costs too much. Overall, this surface mesh is very essential as it lays the groundwork for creating the volume mesh and conducting our CFD analysis.



Surface Mesh Generation and Boundary Identification

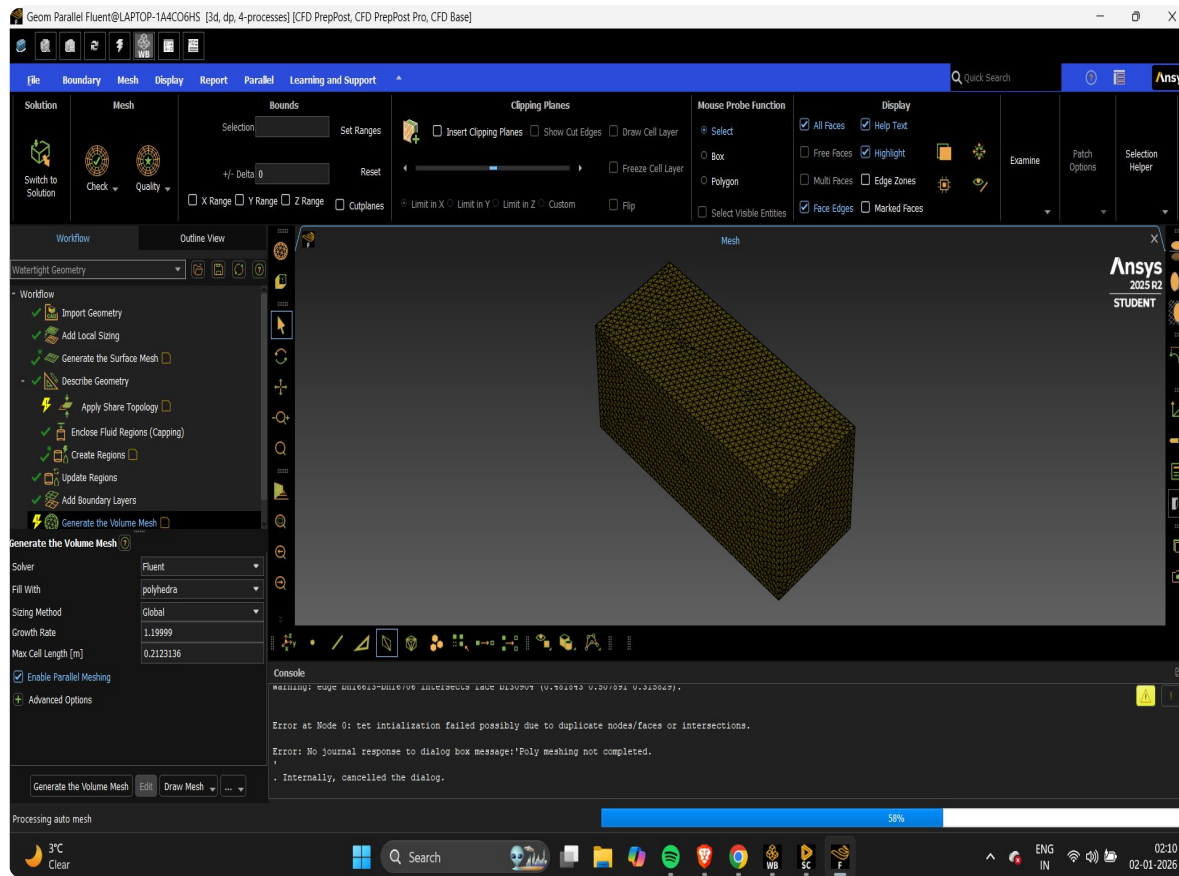


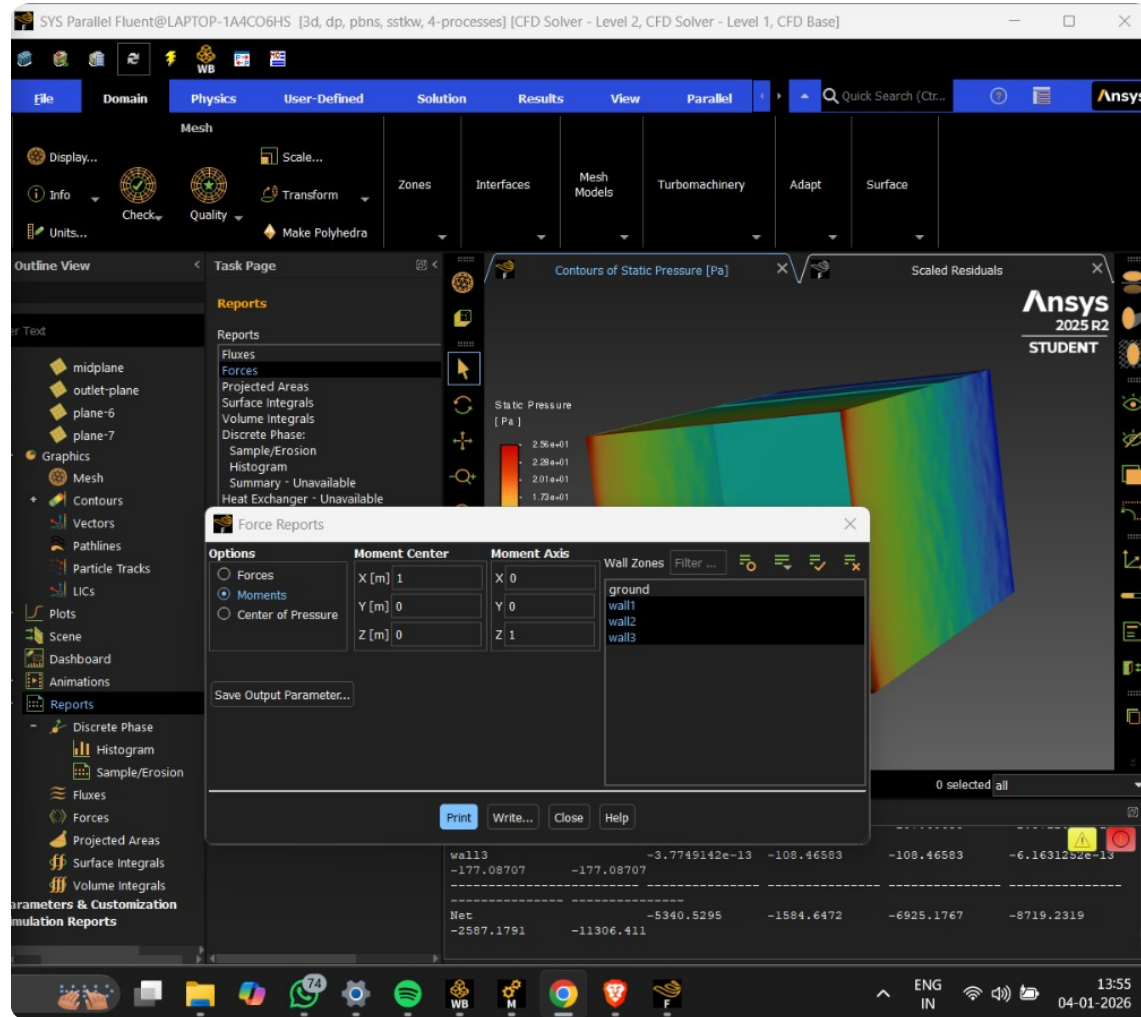
- The figure shows that the surface mesh generation stage in ANSYS Fluent using the Watertight Geometry workflow. Different faces of the rectangular domain are highlighted to identify inlet, outlet, and wall boundaries. This step ensures the correct enclosure of the fluid region before volume meshing. Mesh sizing parameters are applied to control element of distribution and growth rate. Proper boundary identification at this stage is essential for accurate CFD simulations.



Volume Mesh Generation of the Computational Domain

➤ The figure shows the volume mesh generated for the computational domain using ANSYS Fluent Meshing. We opted for a polyhedral mesh to effectively break down the entire fluid volume. The global sizing and growth rate parameters help determine how the cells are distributed throughout the domain. This approach ensures a smooth transition from the surface mesh to the interior cells. Overall, this type of mesh provides reliable flow predictions while keeping computational costs down. When three different fluid with different static properties are subjected to flow through same channel by maintaining the reynold's number constant(25),the flow behavior was quite similar.





Power Calculation from CFD Torque (ANSYS Fluent)

➤ Mechanical power of a rotating body is calculated by using the relation $P = \tau \cdot \omega$, where torque is obtained directly from ANSYS Fluent force reports.

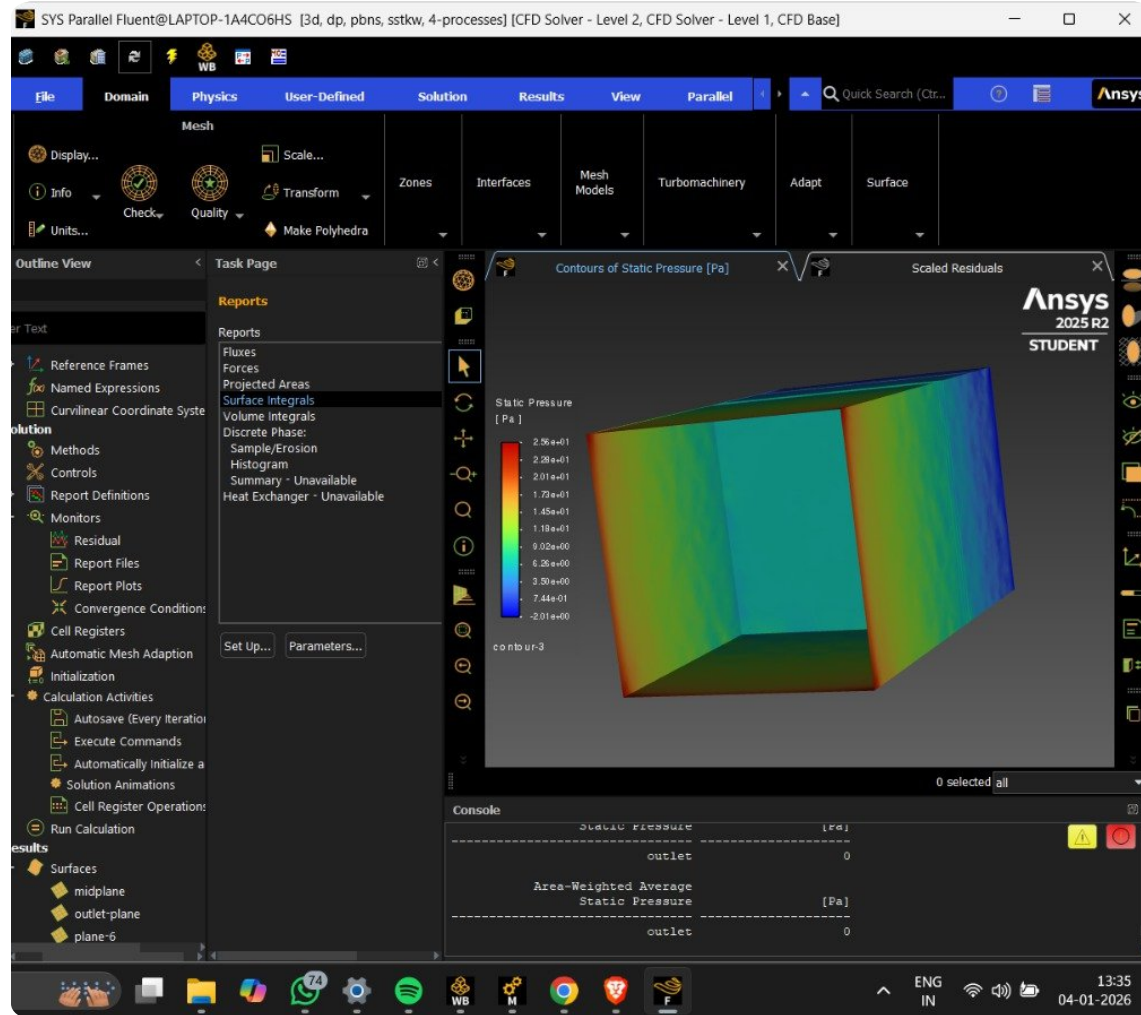
The torque (τ) is measured in the unit $\text{N}\cdot\text{m}$ by selecting the rotating wall surfaces and defining the correct moment center and axis. Angular velocity (ω) must be in **rad/s** and is usually known from the operating condition of the rotor.

If speed is given in RPM, it is converted by using $\omega = 2\pi \times \text{RPM} / 60$.

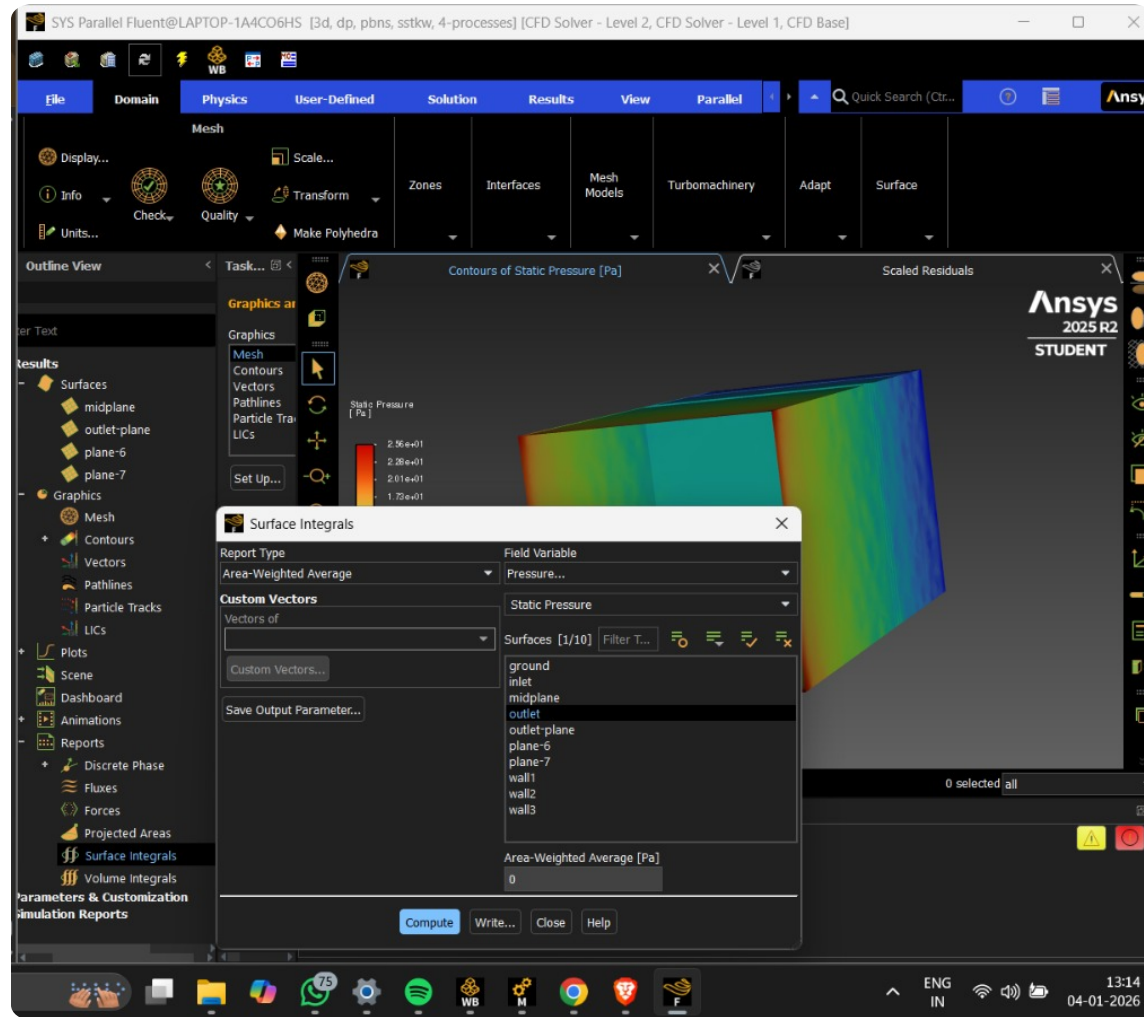
This final power calculation is done **outside Fluent** using the reported torque value.



Pressure Drop Calculation from CFD Results



- The pressure drop across the domain is calculated using $\Delta P = P_{\text{inlet}} - P_{\text{outlet}}$. Area-weighted average static pressure values are extracted at the inlet and outlet planes in ANSYS Fluent. In this case, the inlet pressure is **15.805 Pa** while the outlet pressure is **0 Pa** (reference). Therefore, the resulting pressure drop across the system is **15.805 Pa**. This value represents the total flow resistance and is suitable for direct reporting in the CFD analysis.

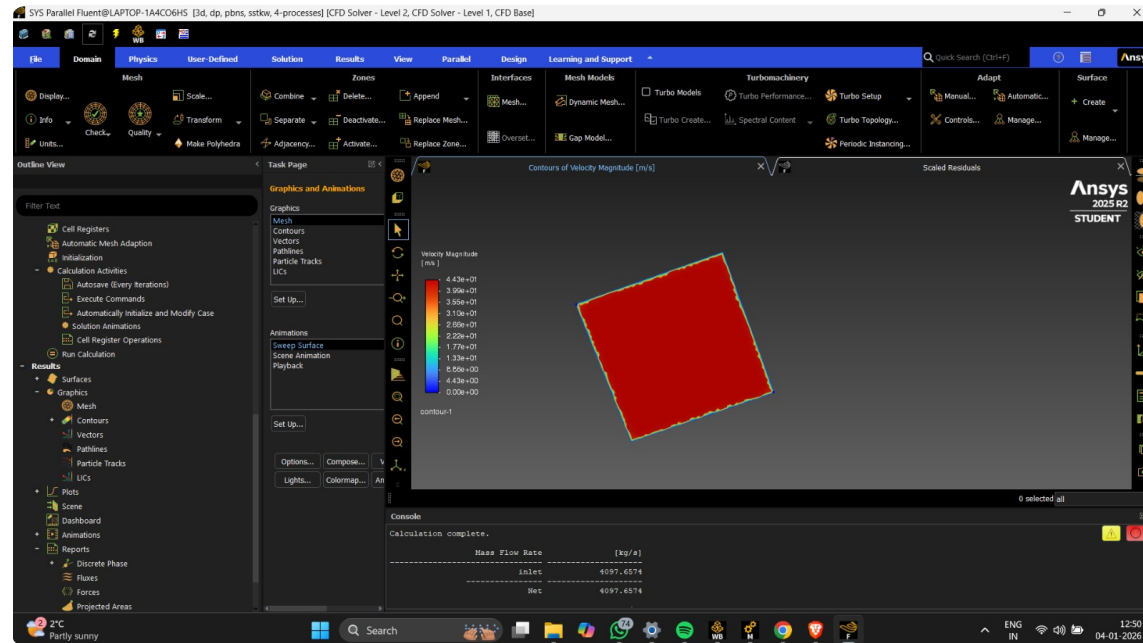


Area-Weighted Average Static Pressure Extraction in ANSYS Fluent

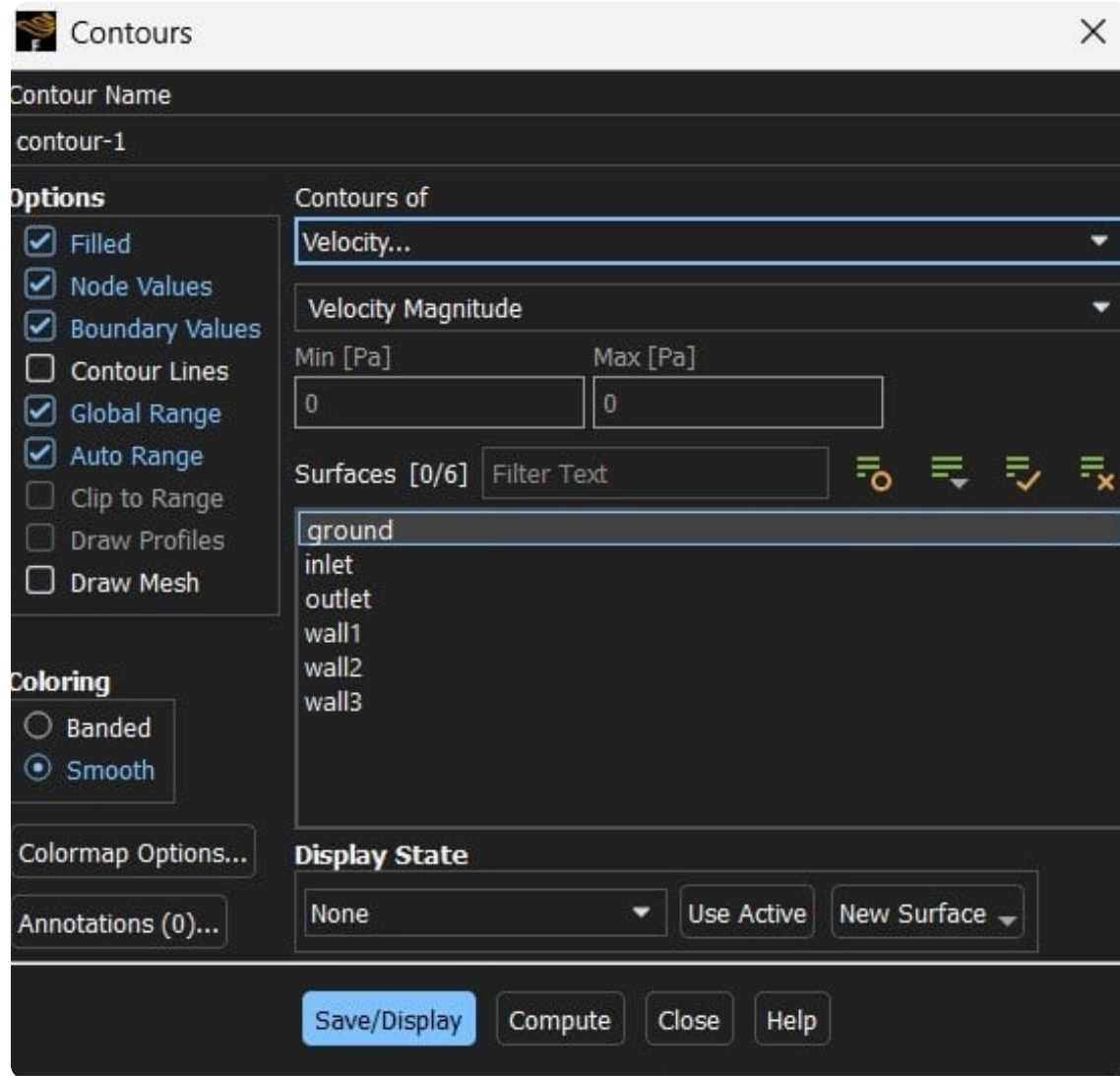
- Area-weighted average static pressure is used to obtain a representative pressure value over a surface. In ANSYS Fluent, this is done via **Reports** → **Surface Integrals** with the report type set to *Area-Weighted Average*. The field variable is selected as **Static Pressure**, and the required surface (inlet or outlet) is chosen. After clicking **Compute**, the pressure value is noted for the selected surface. These inlet and outlet values are later used to calculate the pressure drop across the domain.



Velocity Magnitude Contours in ANSYS Fluent



- The velocity magnitude contour shows the spatial distribution of flow speed within the computational domain. Colors represent velocity levels, where red indicates higher velocity and blue indicates lower velocity regions. This plot helps visualize flow uniformity and identify high-speed or stagnant zones. Contours are generated via
- **Results → Graphics → Contours** by selecting the *Velocity Magnitude*. Such visualization is very essential for the analyzing flow behavior and validating boundary conditions.

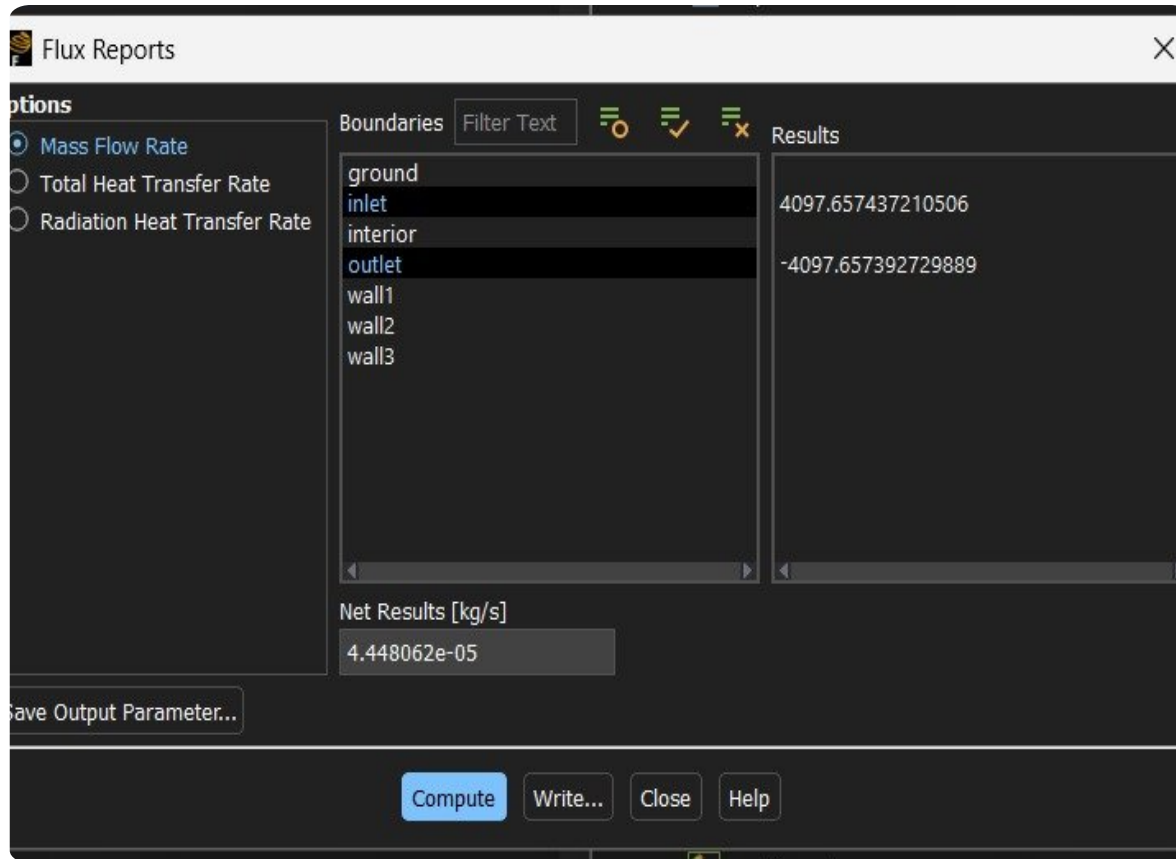


Computational Fluid Dynamics (CFD) Velocity Analysis Interface

➤ This image shows a user-friendly interface for computational fluid dynamics (CFD) or simulation software. It's designed to help users configure and visualize velocity analysis, specifically focusing on "Contours of Velocity Magnitude." You can choose how to display the data with options like "Filled" or "Banded" coloring, and these settings can be applied to various surfaces in your simulation, such as the "ground," "inlet," "outlet," and different "walls." The interface has a sleek dark theme, making it easy for engineers and researchers to customize their visualizations and thoroughly analyze fluid flow data.



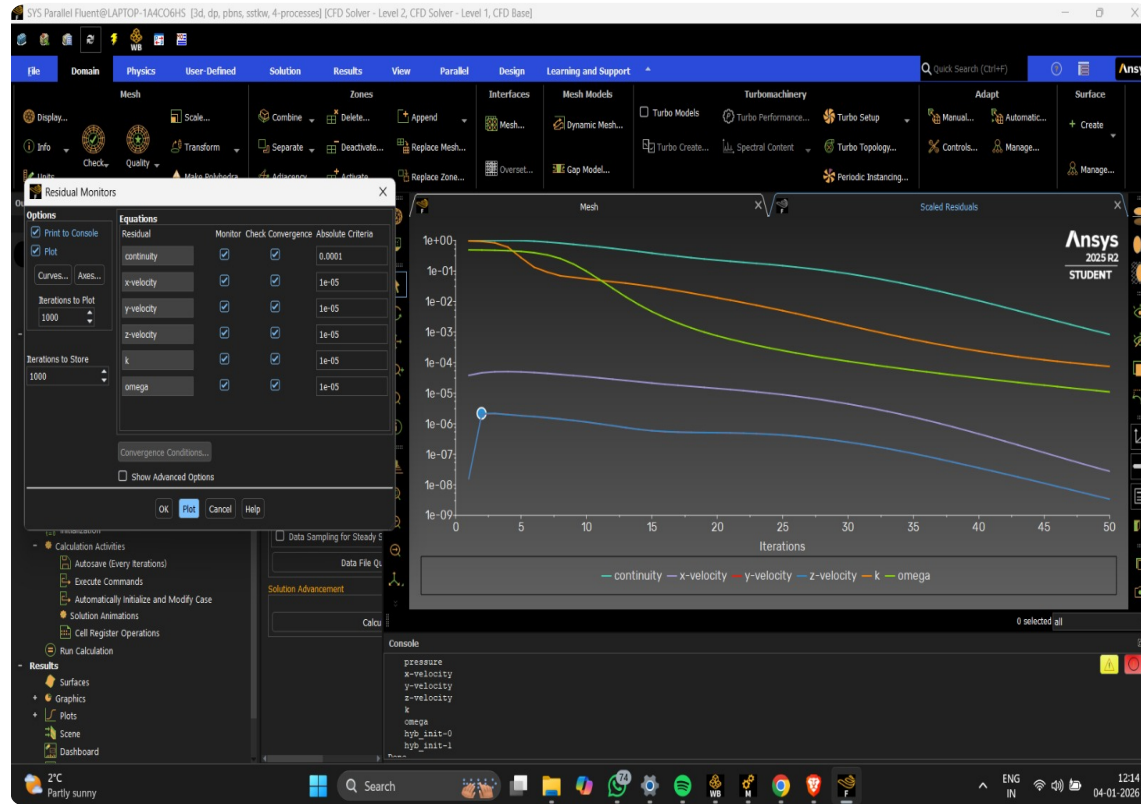
Computational Simulation Flux Analysis



- This image shows the "Flux Reports" interface from simulation software that's likely used for examining fluid dynamics or thermal behavior. It presents calculated flux values, like mass flow rates, across different boundaries such as "ground," "inlet," "outlet," and "walls." Users can see a net flow rate of about 4.448062×10^{-5} kg/s, along with how much each boundary contributes to that overall flow.
- This data is valuable for engineers assessing flow or heat transfer characteristics. The interface is user-friendly and includes options like a "Compute" button to run calculations, along with other handy tools to assist in the analysis.



Residual Convergence History

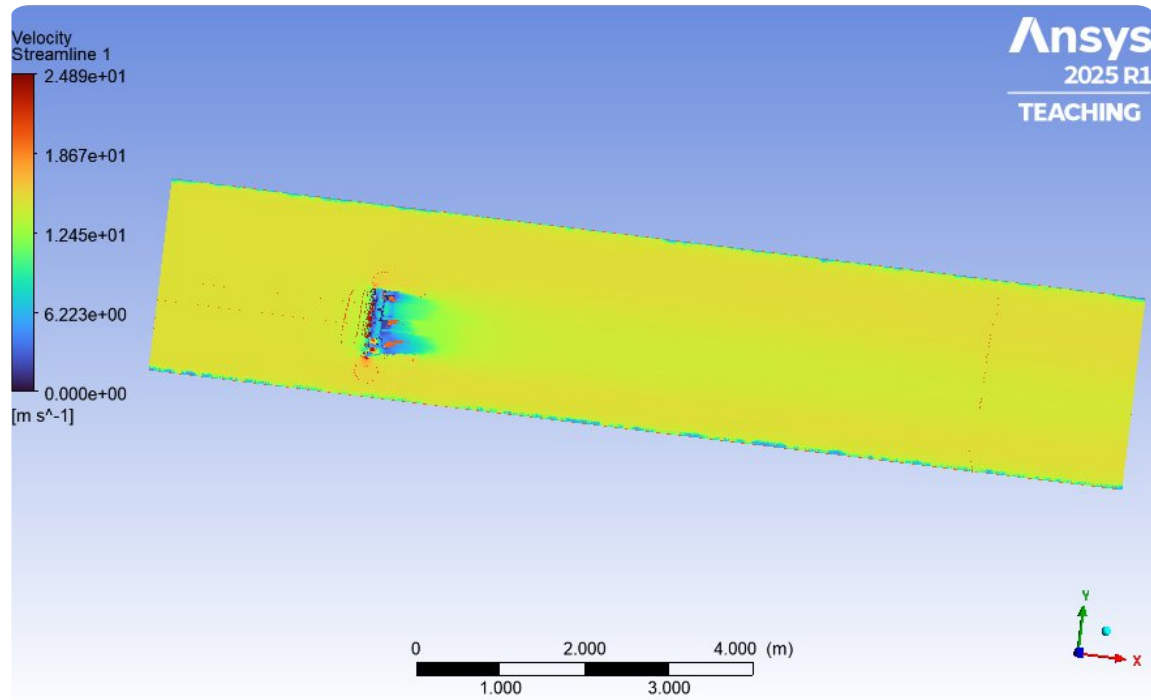


Solution Convergence Monitoring

- The residual plot illustrates the convergence behaviour of the CFD solution over successive iterations.
- Residuals for continuity, velocity components (x, y, z), and turbulence variables (k and ω) decrease steadily with iteration count.
- A reduction of residuals by several orders of magnitude indicates that numerical errors are being progressively minimised.
- After an initial transient phase, the residual curves stabilise and reach a plateau, suggesting convergence of the solution.
- No oscillatory or divergent behaviour is observed, confirming numerical stability of the solver settings.
- The achieved convergence level is suitable for steady state flow analysis and ensures reliability of the extracted CFD results.



Velocity Streamlines



- The velocity streamlines illustrate the flow path and direction of airflow through the computational domain.
- Streamlines accelerate as the flow approaches and passes through the turbine region.
- A low-velocity wake region is observed downstream of the turbine, indicating momentum loss and flow separation.
- The colour variation along the streamlines represents changes in velocity magnitude across the domain.
- Smooth streamline patterns upstream confirm stable and well-developed inlet flow.
- The downstream flow recovery validates the chosen outlet length for capturing wake effects.

Conclusion



- A comprehensive CFD analysis was performed to investigate airflow behaviour around the turbine using ANSYS Fluent.
- Appropriate domain dimensions and mesh refinement ensured fully developed inlet flow, accurate wake capture, and minimal blockage effects.
- Residual convergence and mass flow balance confirmed numerical stability and reliability of the simulation.
- Velocity contours and streamlines revealed realistic flow acceleration, wake formation, and downstream flow recovery.
- Pressure drop and torque values extracted from the converged solution enabled reliable estimation of turbine performance.
- Overall, the CFD methodology adopted in this study is robust and provides a strong foundation for further design optimisation and performance evaluation.

Reference

1. Ren, H.W. et al., 2021. Computational Fluid Dynamics Study of Wake Recovery for Flow Across Hydrokinetic Turbine at Different Depth of Water. *CFD Letters*, 13(2), pp.62–76. [OBJ]
2. Al-Ttowi, A., 2024. Computational Fluid Dynamics (CFD) Investigation of NREL Phase VI Wind Turbine. *Processes*, 12, (accessed 2025). [OBJ]
3. Cioni, S. et al., 2025. Blade-resolved CFD analysis of a floating wind turbine: new insights on unsteady aerodynamics, loads, and wake. *Ocean Engineering*. [OBJ]
4. Carraro, M. et al., 2022. CFD Modeling of Wind Turbine Blades with Eroded Leading Edge. *Fluids*, 7(9), p.302. [OBJ]
5. Menter, F.R., 2015. Best Practice: Scale-Resolving Simulations in ANSYS CFD. ANSYS Inc. [OBJ]
6. Mehr, G. et al., 2017. A Novel Design and Performance Optimization Methodology for Hydraulic Cross-Flow Turbines using Successive Numerical Simulations. SSRN. [OBJ]
7. Betz's law and coefficient of performance, Wikipedia [online]. Available at: https://en.wikipedia.org/wiki/Betz%27s_law (Accessed 2025). [OBJ]